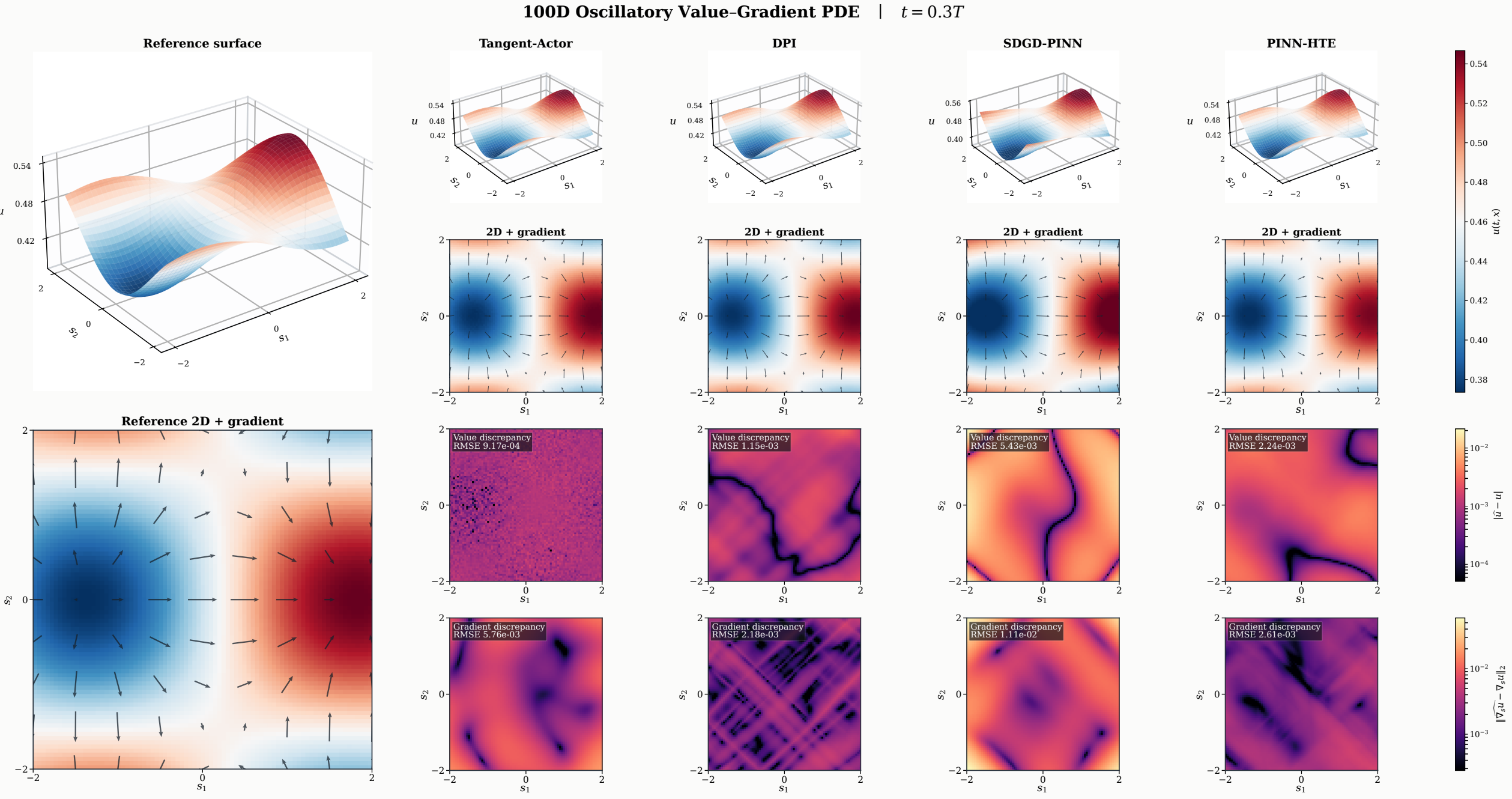

100D Oscillatory Value–Gradient PDE | $t = 0.3T$
Reference surface
Tangent-Actor
DPI
SDGD-PINN
PINN-HTE
2D + gradient
Reference 2D + gradient
Value discrepancy RMSE 9.17e-04
Value discrepancy RMSE 1.15e-03
Value discrepancy RMSE 5.43e-03
Value discrepancy RMSE 2.24e-03
Gradient discrepancy RMSE 5.76e-03
Gradient discrepancy RMSE 2.18e-03
Gradient discrepancy RMSE 1.11e-02
Gradient discrepancy RMSE 2.61e-03
u
s_1
s_2
$u(t, x)$
$|u - \hat{u}|$
$\|\nabla_s u - \widehat{\nabla_s u}\|_2$